1. Replication

 What is it?

Replication means copying data from one PostgreSQL server (**Primary**)
to another (**Replica**) automatically.

```
          INSERT
User --> Primary DB
                           |
                           |
                     Replicate
                           |
                           V
                      Replica DB
```

 Why use it?

* High availability
* Disaster recovery
* Reporting queries on replica
* Load balancing

Types

Streaming Replication

Most common.

Primary > Replica

Changes are continuously sent to replica.

Logical Replication

Replicate selected tables.

```sql
CREATE PUBLICATION sales_pub
FOR TABLE orders;

CREATE SUBSCRIPTION sales_sub
CONNECTION '...'
PUBLICATION sales_pub;
```

Demo Idea

Use two PostgreSQL instances:

* Laptop 1 → Primary
* Laptop 2 → Replica

Insert data into Primary and show it appearing in Replica.

2. Version Control

What is it?

Tracking changes made to database objects over time.

Just like Git tracks source code.

Example

Version 1

sql
CREATE TABLE employees(

```
    id INT PRIMARY KEY,
    name VARCHAR(100)
);
```

Version 2

```sql
ALTER TABLE employees
ADD COLUMN email VARCHAR(100);
```

Version 3

```sql
ALTER TABLE employees
ADD COLUMN salary NUMERIC;
```

 Why use it?

* Track schema changes
* Rollback changes
* Team collaboration
* Deployment automation

Popular Tools

* Flyway
* Liquibase

Demo

Create:

```
text
migrations/
├── V1__create_employee.sql
├── V2__add_email.sql
├── V3__add_salary.sql
```

Show how schema evolves version by version.

3. Backup

What is it?

Creating a copy of database data for recovery.

Logical Backup

Using `pg_dump`

cmd:

```
pg_dump -U postgres companydb > backup.sql
```

Restore:

cmd:

```
psql companydb < backup.sql
```

Custom Backup

bash

```
pg_dump -Fc companydb > backup.dump
```

Restore:

```bash
pg_restore -d companydb backup.dump
```

Physical Backup

Copy PostgreSQL data files.

Tool:

```bash
pg_basebackup
```

Example:

```bash
pg_basebackup \
-D backup_folder \
-U postgres \
```

-P

Real-world Analogy

Concept	Analogy
Replication	Live mirror of your database
Backup	Taking a photo of database at a point in time
Version Control	Tracking every design change to database structure

Interview Difference

Feature	Replication	Backup
Purpose	Availability	Recovery
Real-time	Yes	No
Protects from accidental DELETE	No (DELETE replicates too)	Yes

|
| Disaster Recovery | Partial | Yes
|

Example:

```sql
DELETE FROM employees;
```

If using replication:

```text
Primary  -> DELETE
Replica  -> DELETE
```

Data is gone on both servers.

If you have a backup:

```text
Restore backup
```

Data can be recovered.

One-line summary:

* **Replication** = Keep another server synchronized.
* **Version Control** = Track schema changes.
* **Backup** = Create recoverable copies of data.